ENEL 676 – Distributed Energy Resources

Final Project Overview

Due 10-April-2026

For the final project:

- Individual or up to 3 in a group.
- The project is a DER connection using FortisAlberta standards. Make sure that you mention what version of the standard you are using.
- All calculations can be done in software or by done by hand.
- All deliverables can be made via computer or by hand.
- You will be provided with a high level SLD for a DER connection. You will submit:
    - An updated SLD compliant with FortisAlberta DER interconnection requirements.
    - An Effective Grounding Study.
    - A Short Circuit Study
    - Completed Interconnection Protection Settings and Commissioning sheet (Part 1 only).
    - SCADA data concentrator DNP Points List.
- Final deliverables will be a technical report package submitted on D2L.

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Final Project Briefing

The SLD provided shows a synchronous generator type DER plant that is trying to connect to FortisAlberta's distribution network. The utility has the requirement that the Coefficient of grounding (COG) must be 0.8 or less and the relay de-sensitization must be 10% or less.

Make it so that this DER plant is compliant with the ForitsAlberta requirements at the PCC.

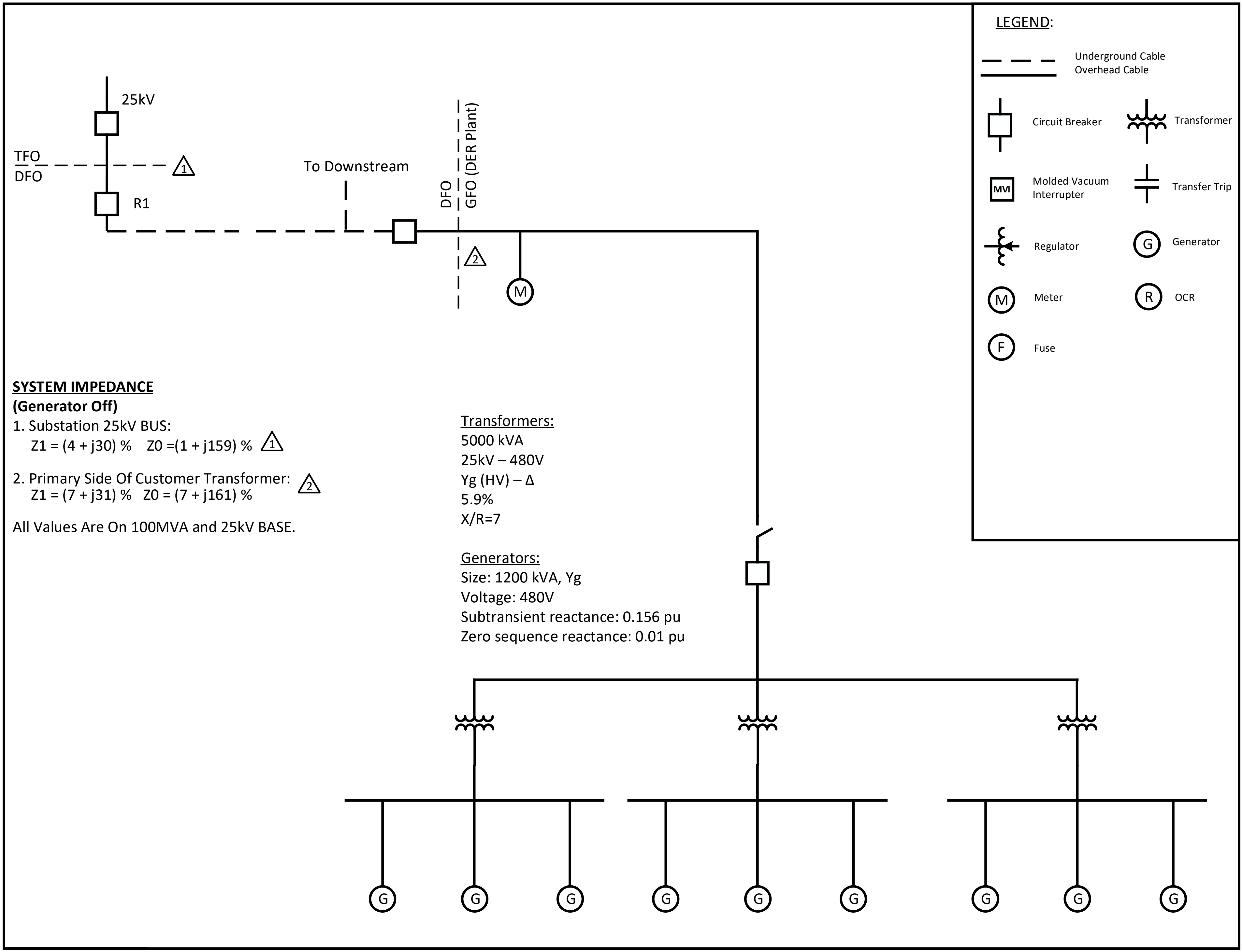
Ensure that you write a small report explaining your findings, how you got them and what the solution is to ensure that the system is compliant with the requirements.

You are required to submit the following:

- An updated SLD compliant with FortisAlberta DER interconnection requirements. Note: you must clearly show any mitigation solutions on the updated SLD. NOTE: the SLD as provided is NOT compliant with FortisAlberta interconnection requirements.
- An Effective Grounding Study. At a minimum, the effective grounding study must provide:
  - The sequence diagrams of the network.
  - The methodology used to determine if the DER is effectively grounded both with and without any mitigation.
  - Calculation of the desensitization index both with and without any mitigation.
  - Clearly outline the solution and include an updated SLD with the solution in it.
- A Short Circuit Study: You are required to determine the short circuit levels at the substation and the PCC. (NOTE: this can be combined with the effective grounding study)
- Completed Interconnection Protection Settings and Commissioning sheet (Part 1 only). Only submit the breaker fail section of the IPSC form (Page 7 of Rev. 09)
- SCADA data concentrator DNP Points List (Note: ignore the full 1547-2018 requirements and only consider the points that FortisAlberta asks for).

FortisAlberta Interconnection standards are here: [Documents and Forms | FortisAlberta]

The applicable standards are DER-02, DER-02A and DER-02B.



SYSTEM IMPEDANCE
(Generator Off)

1. Substation 25kV BUS:
   Z1 = (4 + j30) %   Z0 =(1 + j159) % △1
2. Primary Side Of Customer Transformer: △2
   Z1 = (7 + j31) %   Z0 = (7 + j161) %

All Values Are On 100MVA and 25kV BASE.

Transformers:
5000 kVA
25kV – 480V
Yg (HV) – Δ
5.9%
X/R=7

Generators:
Size: 1200 kVA, Yg
Voltage: 480V
Subtransient reactance: 0.156 pu
Zero sequence reactance: 0.01 pu